TRPC4 subunit identity in the kisspeptin-evoked GnRH-neuron current — Hypothesis Review

Gene: TRPC4 (human, Q9UBN4) · **Hypothesis slug:** kisspeptin-trpc-subunit-identity **Focus:** free_text / core-function & function-assignment reasoning **Seed hypothesis:** Among TRPC1/3/4/5/6/7 co-expressed in GnRH neurons, TRPC4 carries the kisspeptin-evoked cation current *because it alone combines the cSrc-phosphorylated tyrosine and the PIP2-interaction determinants* that the published gating mechanism requires.

Executive Judgment

Verdict: Weakly / partially supported — the specific molecular-determinant rationale is REFUTED.

The residue-level comparison that the hypothesis requests does not do the work the hypothesis claims it does:

1. **cSrc tyrosines (TRPC4 Y959/Y972; Odell 2005,)**

   P 16144838

   are TRPC4-specific — but for the wrong reason. They sit in the hypervariable *distal C-tail* (not conserved in TRPC5/1/3/6/7), and they were characterised as **trafficking / membrane-insertion** determinants in EGF-stimulated COS-7/HEK cells, *not* as gating residues of a kisspeptin current. They are also present in **both TRPC4α and TRPC4β**, so they cannot explain the α-specific identification in GnRH neurons.

2. **The PIP2 determinant does not discriminate TRPC4 from TRPC5.** Phosphoinositide regulation is the *only common regulatory feature of all TRP channels* (Rohacs 2024,).

   P 37871124

   The PIP2-*depletion-activated* behaviour that the kisspeptin mechanism requires (Zhang 2013,

)
P 23744639
is shared by **TRPC5** (Trebak 2009,
)
P 18665391
and is a property of the whole **TRPC1/4/5 subfamily**; the TRPC4-specific part of PIP2 regulation is **isoform-level** (α vs β; Otsuguro 2008,
),
P 18230622
not paralog-level.

3. **The two determinants are not a single coherent "TRPC4-only gating module."** They live in different C-terminal regions and serve different roles (distal-tail trafficking vs α-insert / subfamily PIP2 gating).

4. **Pharmacology points to a heteromer, not a TRPC4 homomer.** The native kisspeptin current is **not blocked by 100 µM La^{3+}** (Zhang 2008,
),
P 18434521
whereas homomeric TRPC4/TRPC5 are *potentiated* by micromolar La^{3+}; native TRPC1/4/5 subunits form PIP2-regulated heteromers (Shi 2012,
).
P 21968068
This favours a heteromeric TRPC1/4(/5) channel.

Net: TRPC4α remains a *plausible contributor* (its mRNA is in GnRH neurons and its PIP2-depletion pharmacology fits), but it is **not uniquely singled out** by the cSrc+PIP2 determinants. The strong "alone combines the determinants" claim should not be used as curation-grade evidence. Most important caveat: subunit identity of the native current was never established functionally (only by blocker pharmacology + TRPC4α mRNA), so any subunit-exclusive statement is under-determined.

Evidence Matrix

#	Citation	Evidence type	Stance	Claim tested	Key finding	Context	Con lim
1	P 16144838 (Odell 2005)	Direct assay + mutant	Qualifies	Identity of cSrc tyrosine	TRPC4 **Y959/ Y972** phosphorylated by Src-family (Fyn); Y959F/ Y972F blocks EGF-induced **membrane insertion** and Ca^{2+} influx	hTRPC4 in COS-7/ HEK293, EGF stimulation	Hig ID; **rol gati** het not
2	Computed (this run)	Structural/ evolutionary	Refutes "alone" basis	Are Y959/Y972 TRPC4-unique and meaningful?	Tyrosines TRPC4-specific but in hypervariable distal tail; absent at aligned position in all 5 paralogs; present in both α and β	UniProt seqs, SW/NW BLOSUM62	Mec (dis alig unc spe gati rele
3	P 23744639 (Zhang 2013)	Direct assay (native)	Supports (partial)	Kisspeptin gating needs PIP2 depletion + cSrc	Kisspeptin→TRPC via PLCβ/**PIP2 hydrolysis** + cSrc; **TRPC4α mRNA** detected (scRT-PCR)	Female mouse GnRH neurons	Hig mec clas **by n onl** sub reso fun
4	P 18434521 (Zhang 2008)	Direct assay (native)	Competing	Subunit identity of current	Kisspeptin NSCC blocked by 2-APB/FFA/ SKF96365/Cd^{2+} but **NOT La^{3+} (100 µM)**	Mouse GnRH neurons	Hig inse arg **het** not hon TRP
5			Qualifies		**TRPC4α (not β)** inhibited by		Hig dete

#	Citation	Evidence type	Stance	Claim tested	Key finding	Context	Con lim
	P 18230622 (Otsuguro 2008)	Direct assay (recomb.)		PIP2 determinant location	PIP2; PIP2 binds α C-terminus; PIP2 breakdown required for activation	Recombinant TRPC4α/β	**α-vs** par excl
6	P 18665391 (Trebak 2009)	Direct assay (recomb.)	Refutes "alone"	Is PIP2-depletion-gating TRPC4-unique?	**TRPC5** also activated by PIP2 depletion & inhibited by PIP2 (not TRPC3/7)	Recombinant TRPC5	Hig **beh TRP** disc
7	P 21968068 (Shi 2012)	Direct assay (native) + interaction	Competing	Heteromer alternative	Native **TRPC1/C5** heteromers; PIP2/PIP3/PKC-activated; TRPC1 confers PI activation	Mouse mesenteric VSMC	Hig hete con diff tiss
8	P 27994151 (Storch 2017)	Interaction + mutant	Refutes "alone"	NHERF/PDZ determinant specificity	**NHERF–TRPC4/5** interaction (C-terminal … VTTRL) required for DAG gating	Recombinant TRPC4/5	Hig NHE dete **sha TRP**
9	P 37871124 (Rohacs 2024)	Review	Refutes "alone"	Is PIP2 interaction discriminating?	Phosphoinositide regulation is the **only common regulatory feature** of TRP channels	Review	Rev pan PIP sing TRP

GO Curation Implications (leads — require curator verification)

- **Do not use the cSrc-tyrosine + PIP2 "combined determinant" argument as evidence for a subunit-exclusive GO Molecular Function assignment to TRPC4.** The determinants are either trafficking-related (Y959/Y972) or shared across the TRPC1/4/5 subfamily / isoform-specific (PIP2), so they do not support a TRPC4-unique gating claim.
- **MF:** Supportable at the level of **GO:0005262 calcium channel activity** / **GO:0005216 monoatomic ion channel activity** / **GO:0005272? (sodium)** — i.e., *non-selective cation channel activity* — for TRPC4 generally; this is retained but is **not diagnostic of the kisspeptin current specifically**. Avoid "protein binding" as a terminal call.
- **BP:** Any term tying TRPC4 to **regulation of GnRH neuron excitability** / **response to kisspeptin** should carry a **context-specific, non-core qualifier** and acknowledge that the native channel is likely **heteromeric** and defined only pharmacologically. Treat as *contributes_to*, not a core exclusive function.
- **CC:** Plasma membrane localization is supported; the α-specific PIP2/NHERF C-terminus links TRPC4α to the **cortical actin** / **NHERF** complex (P 18230622, P 27994151) — a legitimate, better-than-"protein binding" interaction lead if a CC/complex term is desired.
- **Action:** Keep TRPC4's generic cation-channel MF; **do not elevate** "carries the kisspeptin current" to a core, subunit-exclusive function on determinant grounds. Flag TRPC5 and TRPC1/4/5 heteromers as competing annotations.

Mechanistic Scope

- **Immediate molecular activity being tested:** gating of a receptor-operated, PIP2-depletion-sensitive, non-selective cation (Na^+/Ca^{2+}) current downstream of Gαq/PLCβ.
- **Direct gene-product activity:** TRPC4α can form a PIP2-inhibited, PIP2-depletion-activated cation channel (recombinant evidence).

- **Downstream / inferred (not direct determinant evidence):** depolarization and sustained firing of GnRH neurons, GnRH secretion, reproductive-axis activation. These are pathway/physiology outcomes, not proof that TRPC4 homomers carry the current.
- **Trafficking vs gating confound:** the cSrc-Y959/Y972 evidence is about surface delivery of TRPC4, which the hypothesis mis-casts as a gating determinant.

Conflicts and Alternatives

1. **TRPC5 (closest co-expressed paralog):** identical PIP2-depletion gating (P 18665391), shares the …VTTRL NHERF/PDZ motif (P 27994151) → equally qualified on the stated determinants.
2. **Heteromeric TRPC1/4/5 or TRPC1/4:** native heteromers exist and are PIP2-regulated (P 21968068); the La^{3+}-insensitive native current (P 18434521) is more consistent with a heteromer than a homomer.
3. **Isoform confound:** the PIP2 determinant is α-vs-β (P 18230622), while the cSrc tyrosines are α=β — the hypothesis conflates two different "specificities."
4. **mRNA ≠ protein/function:** TRPC4α was identified by single-cell RT-PCR (P 23744639); other subunits are co-expressed, and pharmacology cannot resolve subunit stoichiometry.

Knowledge Gaps

Gap	What was checked	Why it matters	Resolving evidence
Does cSrc phosphorylate Y959/Y972 *in GnRH neurons* and is it needed for the *current* (not trafficking)?	Odell 2005 (heterologous, trafficking)	Hypothesis assumes a gating role	Y959F/Y972F knock-in or dominant-negative in GnRH neurons + kisspeptin recording
Subunit stoichiometry of the native current (homomer vs TRPC1/4/5 heteromer)	Pharmacology + mRNA only	"TRPC4 carries the current" vs heteromer	Subunit-specific KO/ conditional KO in GnRH neurons; concatemers; subunit-specific toxins
Does TRPC5 reproduce the kisspeptin current if TRPC4 removed?	Trebak 2009 shows equivalent PIP2 behavior	Determines whether TRPC4 is necessary	TRPC4 cKO GnRH recording ± TRPC5 compensation
Is the α-specific PIP2 C-terminal insert the actual determinant used in GnRH neurons?	Otsuguro 2008 (recombinant)	Links α isoform to mechanism	Isoform-specific rescue in GnRH neurons

Discriminating Tests (most efficient first)

1. **GnRH-neuron-specific conditional KO of TRPC4 vs TRPC5 vs TRPC1** with kisspeptin whole-cell recording — directly tests necessity and heteromer composition.
2. **Y959F/Y972F TRPC4 knock-in** in GnRH neurons — separates trafficking from gating.
3. **Subunit-selective pharmacology** (e.g., Pico145/HC-070 for TRPC1/4/5; (−)-englerin A sensitivity) on the native kisspeptin current.
4. **Co-IP / proximity ligation** in GnRH neurons for TRPC1/4/5 heteromer detection.
5. **Isoform-resolved expression** (α vs β) at protein level in GnRH neurons.

Curation Leads (require curator verification)

- **Reference to add / verify:**

  P 16144838

  — snippet *"two tyrosine residues in the C terminus of human TRPC4 (hTRPC4), Tyr-959 and Tyr-972, are phosphorylated"* and *"this translocation and association was defective upon mutation of Tyr-959 and Tyr-972"* → supports a **trafficking** role, not gating.
- **Reference to add / verify:**

  P 18230622

  — *"TRPC4alpha but not TRPC4beta was strongly inhibited by intracellularly applied phosphatidylinositol 4,5-bisphosphate (PIP(2))"* → PIP2 determinant is isoform-level.
- **Reference to add / verify:**

  P 18665391

  — *"Depletion of … PIP2 … activates calcium entry and membrane currents in TRPC5-expressing but not in TRPC3- or TRPC7-expressing cells"* → PIP2 gating is shared with TRPC5.
- **Reference to add / verify:**

  P 18434521

  — kisspeptin current *"not by lanthanum (100 microM)"* → heteromer lead.
- **Reference (review orientation):**

  P 37871124

  — PIP2 is the only common TRP regulatory feature.
- **Candidate action change:** downgrade any "TRPC4 uniquely carries the kisspeptin current" statement from core/exclusive to **context-specific contributor**; add competing-subunit note (TRPC5, TRPC1/4/5 heteromer).
- **Candidate GO terms:** retain generic non-selective cation channel MF; add BP *regulation of neuronal action potential / GnRH neuron excitability* as **non-core, context-specific**; do not add determinant-based exclusivity.
- **Suggested curator questions:** Is there any *functional* subunit-resolved evidence (KO, selective blocker) beyond mRNA + blocker pharmacology? Is the α isoform confirmed at protein level?

Limitations of this analysis

- Distal-C-tail alignments are inherently uncertain (low-complexity, variable length across paralogs; TRPC1 is 793 aa so its "last-200" window is not positionally homologous to TRPC4's).
- No GnRH-neuron functional data were (re)generated; all functional claims are from published recordings.
- Precise atomistic PIP2-binding residues were not uniquely defined in the literature, so the PIP2 "determinant" is treated at the region/isoform level rather than as a residue motif.
- Computed provenance: `provenance/trpc_residue_comparison.md` (sequences, methods, residue table).
